

In this page, I investigate some typical finite groups.

Specifically, investigate:

Abelian and Abelianization

Commutator and Derived subgroup

Upper-Central Series and Solvability and Composition Series

Center and Conjugacy classes.

Explicit all subgroup

The general properties of S_n , D_n , C_n will be covered in other page.

$\mathbb{Z}_2 \times S_3$, Conjugacy classes.

Since $|G:G_0| = |C_G|$, find the Centralizer of $(0, (12))$.

Since $(1,2)$ commutes only id and itself, and $\mathbb{Z}_2$ is abelian,

$$C_G(0, (1,2)) = \{(0, (1,2)), (0, \text{id}), (1, (1,2)), (1, \text{id})\}.$$

Therefore, the conjugacy classes of $(0, (1,2))$ has $2 \cdot 6/4 = 3$ elements.

Since $\mathbb{Z}_2$ is abelian (or 0 is identity), the first term never change,

and since the conjugate cannot change the cycle structure,

the conjugacy classes of $(0, (1,2))$ is

$$\{(0, (1,2)), (0, (1,3)), (0, (2,3))\},$$

and similarly, the conjugacy classes of $(1, (1,2))$ is

$$\{(1, (1,2)), (1, (1,3)), (1, (2,3))\},$$

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$$(0, (1,23)) \text{ is } \{(0, (1,23)), (0, (1,32))\}$$

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$$(1, (1,23)) \text{ is } \{(1, (1,23)), (1, (1,32))\}$$

and $(0, \text{id})$ and $(1, \text{id})$ have conjugacy classes only containing itself.

$S_3 \times S_3$, Congruency classes.

Let $(1\ 2) \times (1\ 2) \in S_3 \times S_3$, $(1\ 2)$ commute only id or itself, thus

$$C_G((1\ 2)) = \{(\text{id}, \text{id}), (\text{id}, (1\ 2)), ((1\ 2), \text{id}), ((1\ 2), (1\ 2))\}.$$

So, the congruency classes of $(1\ 2) \times (1\ 2)$ has 4 elements

$\sqrt{S_4}$.

$$\begin{array}{l|l} S_4 = & \{ \text{id}, \\ 1,1,2 & (1\ 2), (1\ 3), (1\ 4), (2\ 3), (2\ 4), (3\ 4), \\ 2,2 & (1\ 2)(3\ 4), (1\ 3)(2\ 4), (1\ 4)(2\ 3), \\ 1,3 & (1\ 2\ 3), (1\ 3\ 2), (1\ 3\ 4), (1\ 4\ 3), (1\ 2\ 4), (1\ 4\ 2), (2\ 3\ 4), (2\ 4\ 3), \\ 4 & (1\ 2\ 3\ 4), (1\ 2\ 4\ 3), (1\ 3\ 2\ 4), (1\ 3\ 4\ 2), (1\ 4\ 2\ 3), (1\ 4\ 3\ 2) \} \\ \text{Partition} & \end{array}$$

A4. Conjugacy classes.

$$A_4 = \{ \text{id}, (12)(34), (13)(24), (14)(23), \\ (123), (132), (124), (142), (134), (143), \\ (234), (243) \}$$

Since the conjugate cannot change the cycle type, the conjugacy class of each element in A_4 are contained the set of same cycle type. But we don't know it exactly equals to the set, let's calculate.

$$(123)(12)(34)(123)^{-1} = (23)(14)$$

$$(124)(12)(34)(124)^{-1} = (24)(31)$$

therefore, the conjugacy classes of $(12)(34)$ equals the set of $(2,2)$ cycles.

$$(132)(123)(132)^{-1} = (312) = (123) \quad (\text{In actually, } (123)^{-1} = (132))$$

$$(124)(123)(124)^{-1} = (243)$$

$$(142)(123)(142)^{-1} = (413) = (134)$$

$$(134)(123)(134)^{-1} = (324) = (243)$$

$$(143)(123)(143)^{-1} = (421) = (142)$$

$$(234)(123)(234)^{-1} = (134)$$

$$(243)(123)(243)^{-1} = (142)$$

$$(12)(34)(123)(12)(34)^{-1} = (214)$$

$$(13)(24)(123)(13)(24)^{-1} = (341) = (134)$$

$$(14)(23)(123)(14)(23)^{-1} = (432) = (243)$$

therefore, the conjugacy classes of (123) has just four elements.

Generally, since $|G:G_\alpha| = C_\alpha$ and $G_\alpha = C_G(\alpha)$ in the conjugate action, therefore number of elements of the orbit is $|G|/|C_G(\alpha)|$.

Since $C_{A_4}((123)) = \{ \text{id}, (123), (132) \},$

the orbit of (123) has $|2/3| = 4$ elements

1A5. The Smallest group of non-abelian Simple group.

Prop. Group of order 60 having more than one Sylow 5-subgroup is simple.

Proof. Using Contradiction. Suppose that there is a normal subgroup $1 < H \triangleleft G$.

By the Sylow Theorem, $n_5 \equiv 1 \pmod{5}$ and $n_5 \mid 12$, so $n_5 \in \{1, 6\}$.

By the hypothesis, n_5 must be 6. Let $P \in \text{Syl}_5(G)$ be a Sylow 5-subgroup.

Since $n_5 = 6$, that is, $|\{gPg^{-1} \mid g \in G\}| = |G|/|N_G(P)| = 6$, therefore $|N_G(P)| = 10$.

Assume that $5 \nmid |H|$. Then, H contains a Sylow 5-subgroup Q of G , because $60 = 2^2 \cdot 3 \cdot 5$, and since H is normal, $gQg^{-1} \subseteq H$ for all $g \in G$, therefore H contains all Sylow 5-subgroups of G .

But, since for any $P_1, P_2 \in \text{Syl}_5(G)$, $P_1 \neq P_2 \Rightarrow P_1 \cap P_2 = \{e\}$, therefore $|H| \geq 5 \cdot 6 - 6 \cdot 1 + 1 = 25$, so the Lagrange's Theorem forces $|H| = 30$.

But, by the Sylow Theorem, $\#\text{Syl}_3(H) \in \{1, 10\}$. If $\#\text{Syl}_3(H) = 1$, then $A \in \text{Syl}_3(H)$ must be a normal subgroup of H , then PA is a group of order $|PA| = |P| \cdot |A| / |P \cap A| = 15/1 = 15$, so $|H|/|PA| = 30/15 = 2$, $PA \triangleleft H$ is normal.

But, order of 5 subgroup in group of 15 is characteristic, because $n_5 \mid 3 \Rightarrow n_5 = 1$, so P characteristic to PA , and $PA \triangleleft H$, so $P \triangleleft H$. It contradicts.

Thus, $\#\text{Syl}_3(H) = 10$, but every non-identity element of Sylow 3-subgroup has order 3, and every non-identity element of Sylow 5-subgroup has order 5, so

$$|H| \geq 6 \cdot 4 + 10 \cdot 2 = 24 + 20 = 44$$

But it cannot be happen, since $|H| = 30$. So, $|H|$ is not divided by 5.

Now, $|H| \in \{2, 4, 6, 12\}$. If $|H| = 6$, then by the Sylow theorem, there is $H_1 < H$ such that $H_1 \in \text{Syl}_3(H)$. If $|H| = 12$, then for $n_3 = \#\text{Syl}_3(H)$, $n_3 = 1$ or $n_3 = 4$. But, $n_3 = 4 \Rightarrow |H| \geq n_3 \cdot (3-1) + 1 + n_2 \cdot (4-1) = 9 + 3 \cdot n_2 \Rightarrow n_2 = 1$

So there is $H_2 < H$ such that H_2 char H . Anyway, H_1, H_2 char H , so $H_1, H_2 \triangleleft G$. Now, $|G/H_1|, |G/H_2| \in \{15, 10\}$, so $|G/H_1| = 12$ has a normal subgroup $K/H_1 \triangleleft G/H_1$. Now, by the corresponding isomorphism theorem, $K \triangleleft G$ and $5 \nmid |K|$, it contradicts by the above argument.

Klein 4-group.

Let $G = \mathbb{Z}_2 \times \mathbb{Z}_2$ be the Klein 4-group.

Then, a map $\varphi: G \rightarrow S_4$:

$$(0,0) \mapsto \text{id}, \quad (1,0) \mapsto (12)(34), \quad (0,1) \mapsto (13)(24), \quad (1,1) \mapsto (14)(23)$$

is Embedding

Let p be a ^{odd} prime number. Let $G = D_{2p} \times C_p$, where D is Dihedral, C is cyclic.

- 1). Find all Sylow subgroups of G .
- 2). Find all p -subgroups.
- 3). Find the Commutator subgroup of G and abelianization.
- 4). Find $\text{Aut}(G)$.

Proof. Since $|G| = |D_{2p}| \cdot |C_p| = 2p^2$, by the Sylow theorem, $n_p = \# \text{Syl}_p(G)$ satisfies that $n_p \equiv 1 \pmod p$ and $n_p | 2$, so that $n_p = 1$. Denote $C_p = \langle x \rangle$. Denote that $D_{2p} = \langle r, s \mid r^p = s^2 = (rs)^2 = e \rangle$. Then, $\langle r \rangle < D_{2p}$ with $|\langle r \rangle| = p$, therefore the subgroup $\langle r \rangle \times C_p < G$ is order of p^2 , so it is the unique Sylow p -subgroup. Meanwhile, by the Sylow theorem again, $n_2 \equiv 1 \pmod 2$ and $n_2 | p^2$, so $n_2 \in \{1, p, p^2\}$. But, since $\langle s \rangle \times \{x^0\}$ and $\langle rs \rangle \times \{x^0\}$ are order 2 subgroups of G , so $n_2 > 1$. If $n_2 = p^2$, then there are p^2 -numbers distinct order 2 elements, but $(a, x^i) \in G$ has order 2 iff $i=0$ and $|a|=2$, but $\{r, r^2, \dots, r^{p-1}\} \subset D_{2p}$ have order of p , so D_{2p} cannot have p^2 -numbers order 2 elements, by just counting. (In fact, $2p < p^2$ also gives same result). So, G has p -numbers Sylow 2-subgroups,

$$\langle rs \rangle \times \{x^0\}, \langle s \rangle \times \{x^0\}, \langle r^2s \rangle \times \{x^0\}, \dots, \langle r^{p-1}s \rangle \times \{x^0\}.$$

2). By the Sylow theorem, all p -subgroups are contained for some Sylow p -subgroup, but since G has a unique Sylow p -subgroup, $\langle r \rangle \times C_p$, so all p -subgroups are $\langle r \rangle \times C_p$ itself and

$$\langle r \rangle \times \{x^0\}, \{e\} \times C_p, \langle (r^k, x^i) \rangle \quad k=1, 2, \dots, p-1,$$

$$(\langle r \rangle \times C_p \cong C_p \times C_p, \text{ it has } p^2 - 1 \text{ elements of order of } p, \dots)$$

3). Since the operation on $G = D_{2p} \times C_p$ was defined by entrywise, so that $[G, G] = [D_{2p}, D_{2p}] \times [C_p, C_p]$. Since C_p is abelian, $[C_p, C_p] = \{x^0\}$, and

$[D_{2p}, D_{2p}] = \langle r \rangle$, so that the commutator is $\langle r \rangle \times \{x^0\}$. Consequently, the abelianization is $D_{2p} \times C_p / \langle r \rangle \times \{x^0\} \cong C_2 \times C_p$.

4).

